The Riemann Hypothesis as a Saturation Law

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Abstract

We exhibit an explicit family of finite Hermitian operators built from Farey fractions for which the Riemann Hypothesis is equivalent to a *lower* bound on a quadratic form, and we prove unconditionally that the critical exponent in that bound is optimal. Let F_N be the Farey fractions of level N (closed convention, both endpoints $0/1$ and $1/1$ included), $m = |F\_N|$, let L_N be the weighted Farey Laplacian with weights $1/(q\_u\ q\_v)^2$, let $\phi_0 = m\hat{}\{-1/2\}\cdot 1$ be its zero mode, and let U be the unitary $\psi(\rho) \mapsto e\hat{}\{2\pi i\rho\}\psi(\rho)$. We prove four results. (1) The exact identity $|\langle\phi_0,\ U\phi_0\rangle|^2 + \|L\_N^+[U,\ L\_N]\phi_0\|^2 = 1$ (Theorem A). (2) The trace evaluation $\langle\phi_0,\ U\phi_0\rangle = (M(N)+1)/m$ with M the Mertens function - the underlying Farey exponential-sum evaluation is classical (Theorem B). (3) Consequently the deficit $D(N) := 1 - \|L\_N^+[U,\ L\_N]\phi_0\|^2$ satisfies the exact formula D(N)\=\((M(N)+1)/m)^2, and RH holds **iff** for every $\varepsilon > 0$, $D(N) = O\_\varepsilon(N\hat{}\{-3+\varepsilon\})$ (Theorem C). (4) Unconditionally, for every $\varepsilon > 0$, $D(N) \geq N\hat{}\{-3-\varepsilon\}$ along infinitely many N - proved self-contained from Hardy's theorem via Landau's lemma - and, importing known explicit oscillation bounds for M, $D(N) \geq c\cdot N\hat{}\{-3\}$ infinitely often for every $c < (1.826054\cdot\pi^2/3)^2 \approx 36.10$ (Theorem D). The critical exponent 3 therefore cannot be improved: RH asserts that the deficit sits exactly at its proven floor. We interpret the identity as a conservation law - a unit of spectral mass split between a flat-mode channel and a commutator (fluctuation) channel - under which RH becomes a saturation principle: maximal transfer of mass from the flat mode into the fluctuation spectrum. The equivalence mechanism is the classical Mertens criterion; the contribution is the operator formulation, the exact identity, and the sharpness theorem. We close with a comparison to the Nyman-Beurling, Báez-Duarte, Li, and Connes reformulations, including Burnol's unconditional lower bound for the Báez-Duarte distance, the closest prior sharpness phenomenon.

Scope and evidence boundary

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Mathematical development assistance: Claude (Anthropic), 2026-06-12 **This paper does not claim a proof of RH.** It proves an exact finite identity, an unconditional sharpness theorem, and an equivalence; the RH-equivalent inequality itself is the open problem. Claims use three evidence classes - **[proved]** (complete proof here or in a cited published source), **[measured]** (finite implementation checks in Appendix A), and **[conjectural]** (open). ## Plain-language orientation The problem is not a lack of equivalent statements of the Riemann Hypothesis. The problem is that many correct equivalences do not expose a finite mechanism that a reader can inspect as a concrete matrix calculation. We therefore begin with an ordinary geometric question: when a unit vector is split into a distinguished constant direction and all directions perpendicular to it, how much of the vector remains in the constant direction, and how much has moved into the perpendicular directions? Linear algebra already answers that question. The two squared components must sum to one. The construction below chooses a finite weighted graph whose vertices are Farey fractions, rotates each vertex by its own complex phase, and uses the graph Laplacian to recover the nonconstant component. The unrecovered amount is not an analogy or a fitted statistic. It is exactly the square of a normalized Mertens sum. Classical number theory then says how quickly that remainder vanishes under the Prime Number Theorem and how quickly it would vanish under the Riemann Hypothesis. Only after that operation is visible do we name it: **saturation** is the approach of the nonconstant channel to its maximal value of one, and the **deficit** is the exact remainder left in the constant channel. The name does not replace the arithmetic. It summarizes an exact projection identity whose rate is governed by the Mertens function. The formulation would fail if any of four checks failed: the universal projection identity, the Farey exponential-sum evaluation, the rescaling from the Mertens function to the deficit, or the cited classical oscillation input. All four are separated below from finite numerical checks. The numerical replay tests implementation only; it supplies no asymptotic evidence and no evidence for RH.

1. Introduction

1.1 Motivation and main results

Most equivalent reformulations of the Riemann Hypothesis place the difficulty in an *upper* bound or a closure statement: a growth bound on an arithmetic sum, the density of a dilation system, the positivity of a derived sequence. Here we present a reformulation in which RH is a **lower bound on an explicit quadratic form attached to a concrete finite operator** - a statement about an explicitly given matrix family, in the territory of spectral-gap, Cheeger, and variational methods - and in which the *sharpness* of that bound is already a theorem.

The construction is elementary to state. For $N \geq 1$ let F_N be the Farey fractions of level N (both endpoints included; see §0 and Appendix A), and let $m = |F_N|$. The weighted Farey Laplacian L_N acts on $\ell^2(F_N)$ with Farey-neighbor adjacency and edge weights $(q_u q_v)^{-2}$; its kernel is spanned by the unit constant vector ϕ_0. Let U be the diagonal unitary $\psi(\rho) \mapsto e^{2\pi i\rho}\psi(\rho)$. Define the **deficit**

$D(N) := 1 - \|L_N^+[U, L_N]\phi_0\|^2$. **[1]**

The main results are:

- **Theorem A (Commutator-Pythagoras identity, Theorem 3.1) [proved].** For any finite-dimensional self-adjoint $L \succeq 0$ with one-dimensional kernel spanned by the unit vector ϕ_0, and any unitary U: $|\langle\phi_0, U\phi_0\rangle|^2 + \|L^+[U, L]\phi_0\|^2 = 1$. **[2]**
- **Theorem B (Mertens trace, Theorem 3.3) [proved; evaluation classical].** For the Farey family, $\langle\phi_0, U\phi_0\rangle = (M(N)+1)/m$, where M is the Mertens function. The underlying Farey exponential-sum evaluation is classical; the operator packaging is what is used here.
- **Theorem C (RH as a lower bound, Corollary 3.5) [proved equivalence].** Exactly, $D(N) = ((M(N)+1)/m)^2$; and RH holds iff for every $\varepsilon > 0$, $D(N) = O_\varepsilon(N^{-3+\varepsilon})$. (The equivalence is proved; the inequality itself is the open problem of §7 and is [conjectural].)
- **Theorem D (unconditional sharpness, Theorems 4.1 and 4.4) [proved].** For every $\varepsilon > 0$, $D(N) \geq N^{-3-\varepsilon}$ for infinitely many N; moreover $D(N) \geq c\cdot N^{-3}$ infinitely often for every $c < (1.826054\cdot\pi^2/3)^2 \approx 36.10$.
- **Theorem E (two-tier saturation, Theorem 3.7) [Tier 1 proved unconditionally; Tier 2 proved equivalence].** The saturation $S(N) = 1 - D(N)$ converges to 1 *unconditionally*, at a rate $D(N) = o(N^{-2})$ that is logically equivalent to the Prime Number Theorem (Tier 1); and $D(N) = O_\varepsilon(N^{-3+\varepsilon})$ iff RH (Tier 2). The framework thus recovers PNT as its leading-order saturation and is equivalent to RH at the sharp rate.

Together, C, D, and E pinch the deficit between an unconditional PNT-strength ceiling ($D(N) = o(N^{-2})$, Tier 1 of Theorem E), an unconditional floor of order N^{-3} infinitely often (Theorem D), and an RH-equivalent ceiling of order $N^{-3+\varepsilon}$ (Theorem C): the critical exponent 3 cannot be improved, the saturation provably occurs at PNT strength with no hypothesis, and RH is precisely the assertion that the deficit achieves its extremal proven order. We call this a *saturation law*. The deficit is exactly $((M(N)+1)/m)^2$, so the

PNT/RH statements are classical facts about M re-expressed as saturation rates; what the operator framework contributes is the *exact identity itself* - a basis-free spectral functional pinned to 1 by unitarity, with $M(N)$ governing only the rate of approach. The reformulation, not the arithmetic, is the contribution; Tier 2 is an equivalence, never a proof of RH. The identity of Theorem A is a conservation law - one unit of spectral mass split between a flat-mode channel and a commutator (fluctuation) channel - and RH becomes the statement that the transfer of mass out of the flat channel is maximal (§5).

The construction requires nothing beyond linear algebra, Ramanujan sums, and one classical oscillation argument (proved self-contained in Lemma 4.2); all analytic difficulty is concentrated in a single inequality, stated as the open problem of §7 together with a graded partial-credit dictionary (Proposition 7.1). §7.3 states the scope and limitations of the contribution explicitly.

1.2 Related work

Classical equivalent reformulations of RH come in several types: growth bounds (Mertens: $M(x) = O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$ [Littlewood 1912; Titchmarsh §14.25]), closure / approximation statements (Nyman-Beurling [Nyman 1950; Beurling 1955]; Báez-Duarte's sequential strengthening [Báez-Duarte 2003]), positivity of derived analytic sequences (Li's criterion [Li 1997; Bombieri-Lagarias 1999]), and trace-formula identities (Connes [Connes 1999]). §6 gives a systematic comparison.

The closest prior sharpness phenomenon is due to Burnol [Burnol 2002], who proved an unconditional lower bound on the Báez-Duarte / Nyman-Beurling distance d_N, controlled by the zeros on the critical line - so the two-sided structure (an unconditional floor whose rate RH asserts is attained) already exists in the Nyman-Beurling framework. The present construction differs in vehicle: it is finite-dimensional at every stage with explicit matrix entries, and it pivots on an *exact* conservation identity rather than an asymptotic closure distance. The distinction from Li's criterion (itself a positivity, i.e. lower-bound-flavored, statement) is the object: a quadratic form of an explicitly given finite operator, versus positivity of a sequence derived analytically from ξ. The arithmetic engine here is the classical Mertens criterion wrapped in an operator identity; we claim a new *formulation* - one that exposes a variational target - not a new equivalence mechanism (see §7.3). The Redheffer-matrix literature is the established genre of "RH as a bound on an explicit finite matrix family"; Remark 6.2 records the differences.

1.3 Organization

§0 fixes notation and the evidence-class convention. §2 defines the operator, its zero mode, and the position unitary, and records the elementary size estimates used later. §3 proves the Commutator-Pythagoras identity (Theorem 3.1), the Mertens trace (Theorem 3.3), and the RH equivalence (Corollary 3.5). §4 proves the unconditional sharpness theorem (Theorem 4.1) with a self-contained oscillation lemma (Lemma 4.2) and an ε-free upgrade with explicit constant (Theorem 4.4). §5 gives the saturation interpretation and proves the spectrum-blindness proposition (Proposition 5.1). §6 compares with other reformulations. §7 states the open problem and three precise questions, proves the partial-credit dictionary (Proposition 7.1), explains why naive attacks fail, and states the limitations of the contribution. Appendix A fixes the level conventions and records the machine verification.

0. Notation and conventions

- $F_N = \{p/q \in [0,1] : \gcd(p,q) = 1,\ 1 \le q \le N\}$, both endpoints $0/1$, $1/1$ included (Appendix A). For $\rho = p/q \in F_N$ in lowest terms, q_ρ (or q_u for a vertex u) denotes its denominator.
- $m = m(N) = |F_N| = 1 + \Sigma_{q\le N} \phi(q)$.

- $M(N) = \Sigma_{q\leq N} \mu(q)$: the Mertens function; for real $x \geq 1$, $M(x) = M(\lfloor x\rfloor)$.
- $c_q(n) = \Sigma_{(a,q)=1} e^{2\pi ian/q}$: the Ramanujan sum; $c_q(1) = \mu(q)$.
- $\langle\cdot,\cdot\rangle$: the standard inner product on $\ell^2(F_N) \cong C^m$, conjugate-linear in the first slot; $\|\cdot\|$ the induced norm.
- L_N: the weighted Farey Laplacian (Definition 2.1); L_N^+ its Moore-Penrose pseudoinverse; $[X, Y] = XY - YX$.
- $D(N) := 1 - \|L_N^+[U, L_N]\phi_0\|^2$: the **deficit** (the central quantity).
- $f = \Omega_\pm(g)$: $\limsup f/g > 0$ and $\liminf f/g < 0$.
- All asymptotic statements are as $N \to \infty$; ε-statements are quantified "for every $\varepsilon > 0$" at the point of use.
- **Evidence classes.** Every claim is one of: **[proved]** - complete proof in this paper or in a cited published source; **[measured]** - output of a finite numerical computation, with the protocol and script identified in Appendix A.3, carrying no asymptotic content; **[conjectural]** - open, asserted by no theorem here. Theorems, lemmas, propositions, and corollaries below are [proved] unless their statements say otherwise; remarks carry explicit labels where they make non-trivial claims.

2. The operator and the zero mode

Definition 2.1 (the weighted Farey Laplacian). Fix $N \geq 1$. The vertex set is F_N (Notation; closed convention). Vertices $u = p/q$, $v = r/s$ (lowest terms) are adjacent, written $u \sim v$, iff $|ps - rq| = 1$ (Farey neighbors). Edge weights: $w_{uv} = (q_u q_v)^{-2} > 0$. The operator $L_N : \ell^2(F_N) \to \ell^2(F_N)$ is

$(L_N \psi)(u) = \Sigma_{v \sim u} w_{uv} (\psi(u) - \psi(v))$. **[3]**

equivalently $L_N = A^\dagger A$ for the weighted oriented incidence map A defined in the companion operator paper [Paper 1, §2]. Everything in this paper uses only two properties of L_N: it is self-adjoint with $L_N \succeq 0$, and $L_N \phi_0 = 0$ for $\phi_0 = m^{-1/2}(1,\ldots,1)$.

For example, F_5 has 11 vertices. The closed convention keeps $0/1$ and $1/1$ as distinct vertices joined by a unit-weight edge. No geometric figure is needed by the proofs; Appendix A fixes the endpoint accounting.

Lemma 2.2 (size of F_N). $m(N) = (3/\pi^2)N^2 + O(N \log N)$. In particular: (a) $m(N) \leq N^2$ for all $N \geq 2$; (b) there exist an absolute constant $c_0 > 0$ and a threshold N_0 such that $m(N) \geq c_0N^2$ for $N \geq N_0$.

Proof. The asymptotic is the classical totient-sum estimate $\Sigma_{q\leq x} \phi(q) = (3/\pi^2)x^2 + O(x \log x)$ [Hardy & Wright, Thm 330]. (a): $m = 1 + \Sigma_{q\leq N} \phi(q) \leq 1 + \Sigma_{q\leq N} q = 1 + N(N+1)/2 \leq N^2$ for $N \geq 2$. (b) is immediate from the asymptotic (any $c_0 < 3/\pi^2$ works for N beyond an effectively computable $N_0(c_0)$). All downstream uses (Corollary 3.5, Theorem 4.1, Theorem 4.4, Proposition 7.1) require only (a), (b), and the asymptotic itself; no claim below depends on a specific numerical value of c_0 or N_0. ■

Remark 2.3 (phases). A magnetic version of L_N, with edge phases θ_{uv} inserted into A, satisfies the same two properties provided the phase field is gauge-trivial on the kernel (i.e. $e^{i\theta_{uv}} = g_u \bar g_v$ for unit scalars g_u, so that the gauged flat vector g_u spans the kernel); see [Paper 1, §2] for the precise condition. We work with the plain (phase-free) weighted Laplacian throughout; no result below uses phases. We also record from [Paper 1] the uniform bound $\|L_N\| \leq 2\pi^2/3$ [proved in Paper 1; not used by any theorem here].

Lemma 2.4 (connectivity, kernel, positivity). For every $N \geq 1$ the Farey graph on F_N is connected. Consequently $L_N \succeq 0$ and $\ker L_N = C\cdot\phi_0$ with $\phi_0 = m^{-1/2}(1,\ldots,1)$.

Proof. *Connectivity.* Let $p/q \in F_N$ with $q \geq 2$. Since $\gcd(p,q) = 1$, there exist integers r, s with $ps - rq = \pm 1$ and $0 \leq r \leq p$, $1 \leq s < q$ (take r/s to be the penultimate convergent of the continued fraction of p/q); then $r/s \in F_N$ (its denominator $s < q \leq N$) and $r/s \sim p/q$. Thus every vertex of denominator ≥ 2 is adjacent to a vertex of strictly smaller denominator; descending, every vertex connects to $\{0/1, 1/1\}$, and $0/1 \sim 1/1$ since $|0\cdot 1 - 1\cdot 1| = 1$. *Positivity and kernel.* For any ψ, $\langle \psi, L_N \psi\rangle = \Sigma_{u\sim v} w_{uv} |\psi(u) - \psi(v)|^2 \geq 0$ (each unordered edge counted once), so $L_N \succeq 0$; equality forces $\psi(u) = \psi(v)$ across every edge, and connectivity forces ψ constant. Hence $\ker L_N = \mathbb{C}\cdot\phi_0$, and ϕ_0 is the unit constant vector. ■

Definition 2.5. $U = \mathrm{diag}(e^{2\pi i\rho})_{\rho \in F_N}$: the position unitary, diagonal in the vertex basis. L_N^+ denotes the Moore-Penrose pseudoinverse of L_N: the inverse of L_N restricted to $(\ker L_N)^\perp = \phi_0^\perp$, extended by zero on $\ker L_N$.

3. The identity

Theorem 3.1 (Commutator-Pythagoras identity). Let L be a self-adjoint positive-semidefinite operator on a finite-dimensional Hilbert space H with $\dim \ker L = 1$, let ϕ_0 be a unit vector spanning $\ker L$, let L^+ be the Moore-Penrose pseudoinverse of L, and let U be any unitary on H. Then

$$|\langle \phi_0, U \phi_0\rangle|^2 + \|L^+ [U, L] \phi_0\|^2 = 1.$$

In particular this holds for (L_N, ϕ_0, U) of §2.

Proof. Let $\{\phi_n\}_{n=0}^{\dim H - 1}$ be an orthonormal eigenbasis, $L\phi_n = \lambda_n\phi_n$, with $\lambda_0 = 0$ and $\lambda_n > 0$ for $n \geq 1$ (such a basis exists since L is self-adjoint with one-dimensional kernel; take $\phi_0 = \phi_0$ up to a phase, which affects nothing below). For $n \geq 1$, using $L\phi_0 = 0$ and self-adjointness of L:

$$\langle \phi_n, [U, L]\phi_0\rangle = \langle \phi_n, U L \phi_0\rangle - \langle \phi_n, L U \phi_0\rangle = 0 - \langle L\phi_n, U\phi_0\rangle = -\lambda_n\langle \phi_n, U\phi_0\rangle.$$

Also $\langle \phi_0, [U, L]\phi_0\rangle = \langle \phi_0, UL\phi_0\rangle - \langle L\phi_0, U\phi_0\rangle = 0$, so $[U, L]\phi_0 \perp \phi_0$ and L^+ acts on it invertibly mode-by-mode:

$$\|L^+[U, L]\phi_0\|^2 = \Sigma_{n\geq 1} \lambda_n^{-2} |\langle \phi_n, [U,L]\phi_0\rangle|^2 = \Sigma_{n\geq 1} \lambda_n^{-2} \cdot \lambda_n^2 |\langle \phi_n, U\phi_0\rangle|^2 = \Sigma_{n\geq 1} |\langle \phi_n, U\phi_0\rangle|^2.$$

Since $U\phi_0$ is a unit vector, Parseval in the basis $\{\phi_n\}$ gives $1 = \|U\phi_0\|^2 = |\langle \phi_0, U\phi_0\rangle|^2 + \Sigma_{n\geq 1}|\langle \phi_n, U\phi_0\rangle|^2$, which is the claim. ■

Remark 3.2 (modularity). Theorem 3.1 is stated for an arbitrary pair (L, U) because its proof uses nothing else: the identity is a bookkeeping consequence of Parseval together with the exact cancellation $\lambda_n^{-2}\cdot\lambda_n^2 = 1$ in each mode. (Indeed U need only be an isometry on the span of ϕ_0; we state it for unitaries since that is the case used.) The Farey structure enters only through Theorem 3.3 below, which computes the flat-channel term arithmetically; the identity converts any arithmetic evaluation of $\langle \phi_0, U\phi_0\rangle$ into an exact statement about a commutator quadratic form. The proof also shows that $\|L^+[U,L]\phi_0\|^2$ depends only on the eigenvector overlaps $|\langle \phi_n, U\phi_0\rangle|^2$, not on the eigenvalues; this *spectrum-blindness* is isolated as Proposition 5.1 and governs the difficulty analysis of §7.

Theorem 3.3 (Mertens trace). With L_N, ϕ_0, U, m as in §2 (closed convention),

$\langle \phi_0, U\phi_0\rangle = (M(N) + 1)/m$, where $M(N) = \Sigma_{q\leq N} \mu(q)$.

Proof. Group the sum by denominator:

$$m\langle \phi_0,U\phi_0\rangle = \Sigma_{\rho\in F_N} e^{2\pi i\rho} = [e^{2\pi i\cdot 0/1} + e^{2\pi i\cdot 1/1}] + \Sigma_{2\leq q\leq N} \Sigma_{1\leq a<q,\ (a,q)=1} e^{2\pi ia/q}.$$

The $q = 1$ bracket contributes $1+1=2=c_1(1)+1$. For each $q \geq 2$ the inner sum is the Ramanujan sum $c_q(1) = \Sigma_{(a,q)=1} e^{2\pi ia/q} = \mu(q)$ [Hardy & Wright, Thm 271]. Hence

$m\langle\phi_0, U\phi_0\rangle = 1 + \Sigma_{q\leq N} c_q(1) = 1 + \Sigma_{q\leq N} \mu(q) = M(N) + 1.$ ■

Remark 3.4 (classical credit and conventions). The Farey exponential-sum evaluation $\Sigma_{\rho\in F_N\cap[0,1)} e^{2\pi i\rho} = M(N)$ is classical (Ramanujan sums: [Hardy & Wright, Ch. XVI]; the Farey-Mertens identity appears explicitly in the literature, e.g. [arXiv:2105.12352]). Theorem 3.3 claims no novelty for this evaluation; the closed convention adds the $+1$ (Appendix A.2). Under the half-open convention the trace is $M(N)$ and nothing downstream changes except additive constants (Appendix A.2).

Corollary 3.5 (RH as a lower bound). Define the deficit $D(N) := 1 - \|L_N^+ [U, L_N]\phi_0\|^2$. Then:

(i) *(exact formula)* $D(N) = |\langle\phi_0, U\phi_0\rangle|^2 = ((M(N)+1)/m)^2$ for every $N \geq 1$.
(ii) *(equivalence)* RH holds **iff** for every $\varepsilon > 0$, $D(N) = O_\varepsilon(N^{-3+\varepsilon})$. **[4]**

Proof. (i) is Theorem 3.1 plus Theorem 3.3 (note $(M(N)+1)/m$ is real). (ii) We use the Mertens criterion: RH $\iff$ for every $\varepsilon > 0$, $M(N) = O_\varepsilon(N^{1/2+\varepsilon})$ [Littlewood 1912; Titchmarsh §14.25].

($\Longleftarrow$) Suppose $D(N) = O_\varepsilon(N^{-3+\varepsilon})$ for every $\varepsilon > 0$. Fix $\varepsilon > 0$. By (i), $D(N)^{1/2} = |M(N)+1|/m$, so by Lemma 2.2(a) ($m \leq N^2$, $N \geq 2$): $|M(N)+1| = m\cdot D(N)^{1/2} \leq N^2 \cdot O(N^{(-3+\varepsilon)/2}) = O(N^{1/2 + \varepsilon/2})$. **[5]** hence $M(N) = O(N^{1/2+\varepsilon/2} + 1) = O(N^{1/2+\varepsilon})$. As ε was arbitrary, RH holds.

($\Longrightarrow$) Suppose RH. Fix $\varepsilon > 0$; then $M(N) = O(N^{1/2+\varepsilon/2})$, so $|M(N)+1| = O(N^{1/2+\varepsilon/2})$. By (i) and Lemma 2.2(b) ($m \geq c_0N^2$, $N \geq N_0$; the finitely many $N < N_0$ are absorbed into the implied constant): $D(N) = ((M(N)+1)/m)^2 = O(N^{1+\varepsilon}/(c_0{}^2N^4)) = O_\varepsilon(N^{-3+\varepsilon})$. ■

Remark 3.6 (machine verification) [measured]. Theorem 3.1 and the integer identity $m\langle\phi_0, U\phi_0\rangle = M(N)+1$ of Theorem 3.3 are machine-verified at $N = 8, 16, 32, 64$ (identity residual at the 10^{-12} level, exact Mertens match); see Appendix A.3 for the verification protocol and script provenance. Both statements are theorems independently of this check; the computation tests the implementation, not the mathematics.

Theorem 3.7 (two-tier saturation). Let $S(N) := \|L_N^+ [U,L_N]\phi_0\|^2$ be the saturation and $D(N)=1-S(N)=((M(N)+1)/m)^2$ its deficit (Corollary 3.5(i)). Then:

(i) *(Tier 1 - unconditional, Prime Number Theorem strength)* $S(N) \to 1$ as $N \to \infty$, unconditionally. Quantitatively $D(N) = o(N^{-2})$, and more precisely $D(N) = O(N^{-2} e^{-c\sqrt{\log N}})$ for an absolute $c > 0$. The deficit rate and the Mertens rate are tied by elementary rescaling, $D(N) = o(N^{-2}) \iff M(N) = o(N)$, and $M(N) = o(N) \iff$ PNT is Landau's classical equivalence (both directions PNT-strength). Hence the rate $D(N) = o(N^{-2})$ is equivalent to the Prime Number Theorem, and holds unconditionally because PNT is a theorem.
(ii) *(Tier 2 - sharp rate, Riemann Hypothesis)* $D(N) = O_\varepsilon(N^{-3+\varepsilon})$ for every $\varepsilon > 0$ **iff** RH holds (Corollary 3.5(ii)).

Proof. (i) By Corollary 3.5(i), $D(N)=((M(N)+1)/m)^2$. The Prime Number Theorem in the form $M(N) = o(N)$ [Landau], together with $m \sim (3/\pi^2)N^2$ (Lemma 2.2), gives $D(N) = (o(N)/m)^2 = o(N^{-2})$; the quantitative form substitutes the de la Vallée-Poussin zero-free-region bound $M(x) = O(x e^{-c\sqrt{\log x}})$, $c > 0$ absolute. Since PNT is a theorem, this is unconditional. The rescaling $D(N) = o(N^{-2}) \iff M(N) = o(N)$ is elementary ($|M(N)+1|/m = o(N^{-1}) \iff M(N) = o(N)$, using $m \asymp N^2$); and $M(N) = o(N) \iff$ PNT is Landau's equivalence [Montgomery-Vaughan, §8.1, p. 244, equation (8.4), chapter DOI 10.1017/CBO9780511618314.010], whose forward direction $M(N)=o(N) \implies$ PNT is Tauberian and of PNT depth (not elementary). The quantitative estimate used above is recorded in

Montgomery-Vaughan, §6.2, p. 182, equation (6.17), chapter DOI 10.1017/CBO9780511618314.008. (ii) is Corollary 3.5(ii). ■

Remark 3.8 (what this does and does not say). The deficit is *exactly* $((M(N)+1)/m)^2$, an explicit algebraic function of the Mertens function, so the *equivalences* in Theorem 3.7 are elementary rescalings of classical facts about M: the framework **reformulates** PNT and RH as saturation rates; it does not re-derive the arithmetic. What is genuinely structural is the underlying identity (Theorem 3.1): a basis-free spectral functional of a concrete self-adjoint operator is pinned to 1 exactly, with the ceiling forced by unitarity and the *rate of approach* governed by $M(N)$. Paired with the unconditional floor of Theorem 4.1 below ($D(N) \geq N^{-3-\varepsilon}$ infinitely often), the deficit is squeezed between an unconditional PNT-strength ceiling (Tier 1) and the proven N^{-3} floor, with RH the single remaining statement about the rate in between. We claim no more: Tier 2 is an equivalence, not a proof, and no part of this recovers RH.

4. Sharpness

Theorem 4.1 (unconditional floor). Let $D(N)$ be the deficit of Corollary 3.5 for the operators of §2. For every $\varepsilon > 0$ there are infinitely many N with

$$D(N) \geq N^{-3-\varepsilon}. \quad [6]$$

and $\langle \phi_0, U\phi_0 \rangle$ changes sign infinitely often. The proof is unconditional; its only nontrivial external input is Hardy's theorem [Hardy 1914] that ζ has a zero on the critical line.

Proof. Hardy's theorem supplies a zero $\rho* = 1/2 + i\gamma$ of ζ with $\gamma \neq 0$. Fix $\varepsilon > 0$ and put $\eta = \min\{\varepsilon,1\}$. Apply Lemma 4.2 below with $\theta = 1/2$ and $\delta = \eta/4 \in (0,1/2)$: there are infinitely many integers N with $M(N) \geq N^{1/2-\eta/4}$ and infinitely many with $M(N) \leq -N^{1/2-\eta/4}$. Fix either subsequence and restrict to N large enough that $N^{1/2-\eta/4} \geq 2$; then $|M(N)+1| \geq |M(N)| - 1 \geq (1/2)N^{1/2-\eta/4}$. By Corollary 3.5(i) and Lemma 2.2(a) ($m \leq N^2$ for $N \geq 2$):

$$D(N) = ((M(N)+1)/m)^2 \geq (1/4)\cdot N^{1-\eta/2}/N^4 = (1/4)\cdot N^{-3-\eta/2}. \quad [7]$$

For $N \geq 4^{2/\eta}$ we have $N^{\eta/2} \geq 4$, hence $(1/4)\cdot N^{-3-\eta/2} \geq N^{-3-\eta} \geq N^{-3-\varepsilon}$, giving the stated bound along infinitely many N. Sign changes: along the two subsequences $M(N)+1$ is ≥ 1 and ≤ -1 respectively, so $\langle \phi_0, U\phi_0 \rangle = (M(N)+1)/m$ (with $m > 0$) takes both signs infinitely often. ■

Lemma 4.2 (oscillation; classical, proof included). Suppose $\zeta(\rho*) = 0$ with $\mathrm{Re}\ \rho* = \theta \in (0,1)$ and $\mathrm{Im}\ \rho* \neq 0$. Then for every $\delta \in (0, \theta)$,

$$M(x) = \Omega_\pm(x^{\theta-\delta});$$

that is, neither $M(x) \leq x^{\theta-\delta}$ for all large x nor $M(x) \geq -x^{\theta-\delta}$ for all large x.

Proof. We use the Mellin representation, valid for $\mathrm{Re}\ s > 1$:

$$1/\zeta(s) = \Sigma_{n\geq 1} \mu(n)\ n^{-s} = s \int_1^\infty M(x)\ x^{-s-1}\ dx. \quad [8]$$

(partial summation; absolute convergence for $\mathrm{Re}\ s > 1$ from the trivial $|M(x)| \leq x$). We also use Landau's theorem on integrals with eventually sign-constant integrand: *if $h : [1,\infty) \to R$ is locally integrable, $h(x) \geq 0$ for all $x \geq x_0$, and $G(s) = \int_1^\infty h(x)x^{-s-1}dx$ has abscissa of convergence $\sigma_c > -\infty$, then G has a singularity at the real point $s = \sigma_c$.* This is exactly Montgomery-Vaughan, Lemma 15.1, p. 463, §15.1, chapter DOI 10.1017/CBO9780511618314.017; the authors state the eventual-nonnegative integral form and derive it from their Dirichlet-series Landau theorem. The proof below uses only that stated form. The standard statement assumes $h \geq 0$ on the whole ray; the eventual version follows by splitting $G = \int_1^{x_0} + \int_{x_0}^\infty$, since the first term is entire in s and singularities and abscissae are unaffected by adding an entire function.

Suppose, for contradiction, that $M(x) \le x^{\theta-\delta}$ for all $x \ge x_0$. Set $h(x) = x^{\theta-\delta} - M(x) \ge 0$ for $x \ge x_0$, and

$G(s) = \int_1^\infty h(x)\, x^{-s-1}\, dx = 1/(s - \theta + \delta) - (1/s)\cdot(1/\zeta(s))$. **[9]** for $\mathrm{Re}\ s > 1$,

by [8] and $\int_1^\infty x^{\theta-\delta-s-1}dx = 1/(s-\theta+\delta)$. The right-hand side is meromorphic on $\mathrm{Re}\ s > 0$ with poles only at $s = \theta - \delta$ and at zeros of ζ. On the real segment $(\theta - \delta, \infty)$ it is analytic: $\zeta(s) \neq 0$ for real $s > 0$ ($\zeta(s) > 0$ for $s > 1$; $\zeta(s) < 0$ for $0 < s < 1$, e.g. by $(1 - 2^{1-s})\zeta(s) = \Sigma\ (-1)^{n-1} n^{-s} > 0$ and $1 - 2^{1-s} < 0$ there; $s = 1$ is a pole of ζ, so $1/\zeta$ is analytic and vanishing there, and $\theta - \delta > 0$ keeps the segment inside $\mathrm{Re}\ s > 0$), and the only real pole, $s = \theta - \delta$, lies at the left endpoint. Let σ_c be the abscissa of convergence of G. If $\sigma_c = -\infty$ the integral converges everywhere and G is entire; otherwise, by Landau's theorem, G is singular at the real point σ_c, and since G is analytic on $(\theta - \delta, \infty)$, this forces $\sigma_c \le \theta - \delta$. In either case the integral defining G converges, and G is analytic, on the open half-plane $\mathrm{Re}\ s > \theta - \delta$ - which contains $\rho* = \theta + i\gamma$, where $1/(s-\theta+\delta) - G(s) = (1/s)/\zeta(s)$ has a pole (a zero of ζ at $\rho*$ makes $1/\zeta$ singular there, while $1/(s-\theta+\delta)$ and $1/s$ are analytic at $\rho*$). Contradiction. Hence $M(x) > x^{\theta-\delta}$ for arbitrarily large x. Applying the same argument to $h(x) = x^{\theta-\delta} + M(x)$ (i.e. to $-M$) gives $M(x) < -x^{\theta-\delta}$ for arbitrarily large x. ■

Remark 4.3 (provenance). Lemma 4.2 is classical - it is the standard Landau-lemma transposition, from $\psi(x) - x$ to $M(x)$, of the oscillation theorems of Landau and Ingham ([Ingham 1942]; [Titchmarsh §14]; [Montgomery-Vaughan, Lemma 15.1 and §15.1]). We include the proof to keep the paper self-contained: with it, the only external inputs to Theorem 4.1 are Hardy's theorem and the cited textbook lemma on sign-constant Dirichlet integrals.

Theorem 4.4 (ε-free floor with explicit constant). Unconditionally, for every constant $c < (1.826054\cdot\pi^2/3)^2 = 36.10\ldots$, there are infinitely many N with

$D(N) \ge c\cdot N^{-3}$. **[10]**

Proof. Hurst [2018, DOI 10.1090/mcom/3275], building on the Ingham-Odlyzko-te Riele computational framework, proves unconditionally $\limsup_{x\to\infty} M(x)/\sqrt{x} \ge 1.826054$ and $\liminf_{x\to\infty} M(x)/\sqrt{x} \le -1.837625$. Best and Trudgian [2015] established the earlier symmetric magnitude 1.6383; that value is not the source of the constant used here. Since $M(x) = M(\lfloor x\rfloor)$ and $\sqrt{x} \ge \sqrt{\lfloor x\rfloor}$, the limsup is attained along integers: for any $\kappa < 1.826054$ there are infinitely many integers N with $M(N) \ge \kappa\sqrt{N}$, and along that subsequence $M(N)+1 \ge \kappa\sqrt{N}$. By Corollary 3.5(i) and Lemma 2.2's asymptotic $m = (3/\pi^2)N^2(1 + o(1))$:

$$D(N) = ((M(N)+1)/m)^2 \ge \kappa^2 N / ((3/\pi^2)^2 N^4(1+o(1))) = (\kappa\pi^2/3)^2\cdot N^{-3}\cdot(1+o(1)).$$

Given $c < (1.826054\cdot\pi^2/3)^2$, choose $\kappa < 1.826054$ with $(\kappa\pi^2/3)^2 > c$; the $(1+o(1))$ factor exceeds $c/(\kappa\pi^2/3)^2$ for all large N, proving the claim. ■

Remark 4.5 (the two floors; falsifiability; Mertens-conjecture context). Theorem 4.1's proof is self-contained given Hardy's theorem and carries the logical structure of the paper; Theorem 4.4 imports deeper explicit oscillation bounds but eliminates the ε and supplies the constant. Together with Corollary 3.5 they pinch the deficit between an unconditional floor ($\ge cN^{-3}$ infinitely often, every $c < 36.10\ldots$) and the RH-equivalent ceiling ($O(N^{-3+\varepsilon})$ for every $\varepsilon > 0$): RH is the statement that the deficit achieves its extremal proven order - a *saturation law*. Two contextual notes. (i) Falsifiability: any numerical claim that $D(N)$ decays strictly faster than N^{-3} for all large N would contradict Theorem 4.4 - a built-in sanity check for computations. (ii) The disproof of the Mertens conjecture [Odlyzko-te Riele 1985] shows limsup

$M(x)/\sqrt{x} > 1$, so the floor is not an artifact of small constants: the deficit provably returns to the N^{-3} scale with constant exceeding $(\pi^2/3)^2 \approx 10.8$ infinitely often.

5. The saturation interpretation

Theorem 3.1 is a conservation law: the unit vector $U\phi_0$ distributes one unit of mass between the flat channel ($|\langle\phi_0, U\phi_0\rangle|^2$) and the fluctuation channel ($\Sigma_{n\geq 1} |\langle\phi_n, U\phi_0\rangle|^2$). We record the structural point as a proposition.

Proposition 5.1 (spectrum-blindness). Let (L, ϕ_0, U) be as in Theorem 3.1 and let $L' = f(L)$ for any function f with $f(0) = 0$ and $f(\lambda_n) > 0$ for all $n \geq 1$ (so L' is self-adjoint, $\succeq 0$, with the same kernel and the same eigenvectors). Then

$\|L'^{+}[U, L']\phi_0\|^2 = \|L^{+}[U, L]\phi_0\|^2$. **[11]**

In particular the deficit $D(N)$ is invariant under every such respacing of the spectrum of L_N, and depends only on the eigenvector overlaps $\{|\langle\phi_n, U\phi_0\rangle|^2\}_{n\geq 1}$.

Proof. By the computation in Theorem 3.1's proof, both sides equal $\Sigma_{n\geq 1} |\langle\phi_n, U\phi_0\rangle|^2$. ■

In the fluctuation channel, the commutator weights mode n by λ_n^2, which L_N^+ removes exactly - Proposition 5.1 is the precise statement that the deficit is blind to the spectrum and sees only the overlap-mass distribution. (This is also why the open problem of §7 is hard: spectral-gap information alone cannot move the deficit.) In physics-free terms, the identity is a finite analogue of a sum rule, and RH is the statement that the flat channel empties at the fastest rate not excluded by the oscillation theorem of §4 - equivalently, the commutator channel absorbs the maximal share compatible with Theorem 4.1.

The identity is therefore a universal projection formula, not an operator-specific spectral mechanism. The Farey operator supplies a concrete carrier, while all arithmetic content enters through the flat-mode trace. Neither a spectral gap nor eigenvalue localization controls $D(N)$ after the exact cancellation in Proposition 5.1.

Remark 5.2 (source framework; no physical claims) [conjectural as physics; no mathematical content]. In the source physical framework (Super Information Theory, where L_N arises as a coherence-transport operator; see the companion Papers 2 and 6), the saturation law reads: *the Riemann Hypothesis is a principle of maximal coherence transfer*. We record this interpretation without physical claims; no theorem in this paper depends on this section, and this section may be excised without affecting any proof.

6. Comparison

Criterion	Type	Object
Nyman-Beurling / Báez-Duarte	closure / approximation	$L^2(0,1)$ dilation system
Burnol 2002	unconditional lower bound on the BD distance	$L^2(0,1)$ dilation system
Li	positivity of a derived sequence	λ_n from ξ
Connes	trace formula identity	adèle class space
Mertens (Littlewood)	growth bound	$M(x)$
This paper	**lower bound on a quadratic form**	finite Farey operator, explicit

Distinctive features: finite-dimensional at every stage with explicit entries; the RH-statement is variational; the sharpness half is already a theorem (Theorems 4.1, 4.4). The structural relative of that last feature is Burnol's unconditional lower bound $\text{liminf\_\{}\lambda\to0\text{\} d(}\lambda\text{)}\cdot\text{(log(1/}\lambda\text{))\^{}\{1/2\}} \geq (\Sigma\_\rho\ 1/|\rho|^2)\text{\^{}\{1/2\}}$ for the Nyman-Beurling/Báez-Duarte distance [Burnol 2002]: there too the RH rate is a proven floor. The differences are the vehicle - an asymptotic Hilbert-space distance there, an exact finite identity with explicit matrix entries here - and the driver of the floor (the critical-line zeros directly there; the Mertens oscillation here).

Remark 6.1 (not Hilbert-Pólya). This paper is not a Hilbert-Pólya proposal. The eigenvalues of L_N are not the zeros of ζ; indeed by Proposition 5.1 the RH-equivalent quantity is invariant under arbitrary respacings of the spectrum. RH enters through a distinguished *vector* ($U\phi_0$ against the spectral decomposition), not through the spectrum.

Remark 6.2 (matrix-genre relatives). The Redheffer-matrix literature [Redheffer 1977; arXiv:0811.3701] is the established genre of "RH as a norm / eigenvalue bound on an explicit finite matrix family" (det\A_n\=\ M(n)). The present family differs in carrier (Farey graph vs divisibility matrix) and in that the core statement is an exact identity rather than a norm conjecture.

Remark 6.3 (companion work; imports flagged). Related Branch C papers are deposited separately: Paper 1 (10.5281/zenodo.21231873), Paper 2 (10.5281/zenodo.21231875), Paper 4 (10.5281/zenodo.21231879), Paper 5 (10.5281/zenodo.21231881), Paper 7 (10.5281/zenodo.21231887), and Paper 8 (10.5281/zenodo .21231891). Paper 1 contains the same universal projection identity. The corrected BC-045 route supports only an unspecified lower bound $\lambda_2 \geq c/(N^4 \log N)$ with absolute c>0; it does not supply an explicit ln(2)/4 constant. None of this paper's theorems depends on a companion result, a spectral-gap estimate, a localization measurement, or an infinite-operator bridge.

7. Open problem, questions, and limitations

7.1 The open problem and the partial-credit dictionary

Open Problem [conjectural]. Prove: for every $\varepsilon > 0$, $D(N) = 1 - \|L\_N^{+}[U,L\_N]\phi_0\|^2 = O(N\text{\^{}\{-3+}\varepsilon\text{\}})$. By Corollary 3.5 this is equivalent to the Riemann Hypothesis, and it remains open here.

Proposition 7.1 (partial-credit dictionary). For each fixed $\delta \in [2, 3]$ the following are equivalent:

(i) for every $\varepsilon > 0$, $D(N) = O\_\varepsilon(N\text{\^{}\{-}\delta+\varepsilon\text{\}})$;
(ii) for every $\varepsilon > 0$, $M(N) = O\_\varepsilon(N\text{\^{}\{2-}\delta/2+\varepsilon\text{\}})$.

Proof. Immediate from the exact formula D(N)\=\((M(N)+1)/m)² (Corollary 3.5(i)) and Lemma 2.2(a),(b), exactly as in the proof of Corollary 3.5(ii) with 1/2 replaced by $2 - \delta/2$ (for $\delta \in [2,3]$ the exponent $2 - \delta/2 \in [1/2, 1]$ stays positive, so the ±1 and small-N absorptions are identical). ■

At $\delta = 2$, (ii) is the trivial bound $M(N) = O(N)$; at $\delta = 3$ it is RH. Hence any proved exponent $\delta > 2$ is new information about Mertens growth, and the problem admits graded, falsifiable intermediate targets rather than an all-or-nothing jump. (We note the converse reading: by the same dictionary, progress on D(N) *is* progress on M(N) and is exactly as hard - the formulation reallocates the difficulty, it does not reduce it.)

7.2 Why the obvious approach fails, and three questions

By Proposition 5.1 the deficit is spectrum-blind: no bound on the spectral gap of L_N can, by itself, bound D(N) - the deficit depends only on how the overlap mass $|\langle\phi\_n,U\phi_0\rangle|^2$ distributes over modes, not on where the eigenvalues sit. The companion obstruction paper (Paper 4) proves no-go theorems eliminating the global-trace and shell-diagonal bridge shapes. A successful operator-side attack would therefore need new control of the overlap-mass distribution itself; no such control is proved or measured in this revision package.

We close with three precisely posed questions, in increasing strength:

Question 1 (any nontrivial exponent) [conjectural]. Does there exist $\delta > 2$ such that $D(N) = O_\varepsilon(N^{-\delta+\varepsilon})$ for every $\varepsilon > 0$? By Proposition 7.1 any such δ gives $M(N) = O_\varepsilon(N^{2-\delta/2+\varepsilon})$, a zero-free-region-grade statement; even $\delta = 2 + \eta$ for a fixed $\eta > 0$ would be of interest.

Question 2 (overlap-mass localization) [conjectural]. Prove any nontrivial upper bound on $\Sigma_{n \in S} |\langle \phi_n, U\phi_0 \rangle|^2$ for an explicitly described set S of modes (e.g. modes localized at convergents of bounded depth), sharp enough to feed attack (a) below. No theorem or governed measurement about this distribution is supplied here.

Question 3 (interpolation) [conjectural]. Construct an analytic family interpolating between this paper's identity at the $s = 1$ normalization and the scattering-ratio identity at $s = 1/2$ (companion Paper 7), with a maximum-principle or monotonicity statement strong enough to transfer information between the endpoints.

Attacks. The proven structure suggests two: (a) spectral-gap / Cheeger control of the *commutator expansion* - not of the deficit directly - using the gap bracket and the localization of low modes on deep convergents (inputs from Paper 5); (b) the interpolation of Question 3, developed in Paper 7 (*Eisenstein-cusp pairing*), with operator inputs from Paper 5. Separately, the identity side of this paper is being formalized in Lean (bridge chain BC-014…BC-041), so a future proof of the inequality would land on a machine-checked foundation.

7.3 Scope and limitations

Three limitations should be stated plainly. **(1) The equivalence mechanism is classical.** Corollary 3.5 routes through the Mertens criterion [Littlewood 1912]; this paper proves no new bound on $M(N)$, and by Proposition 7.1 any progress on the deficit is *identical* in strength to progress on Mertens growth. The contribution is the operator formulation (an exact finite identity exposing a variational target) and the sharpness theorem, not a new route to RH. **(2) The sharpness floor is along a subsequence.** Theorems 4.1 and 4.4 give $D(N) \geq cN^{-3}$ *infinitely often*, not for all N; indeed $D(N)$ vanishes whenever $M(N) = -1$, which happens infinitely often if M changes sign through -1 (and is [measured] to happen at small N). The saturation statement is about the extremal order, not a pointwise floor. **(3) Numerical support is finite and small.** Machine verification (Appendix A.3) uses dense pseudoinverse checks only at $N = 8,16$ and a pseudoinverse-free projection / trace check through $N = 512$. It checks an implementation of proved identities and supports no asymptotic inference. No claim in this paper depends on numerics; conversely, the numerics establish nothing beyond the stated finite levels.

Appendix A: level conventions and machine verification

A.1 Convention. Throughout this paper $F_N = \{p/q \in [0,1] : \gcd(p,q) = 1,\ 1 \leq q \leq N\}$ with **both endpoints included**: $0/1$ and $1/1$ are distinct vertices. Then $m = |F_N| = 1 + \Sigma_{q \leq N} \phi(q)$ (the $+1$ accounts for the second denominator-1 vertex). The edge $0/1 - 1/1$ is present at every level ($|0 \cdot 1 - 1 \cdot 1| = 1$), and Lemma 2.4's connectivity argument is unaffected by the convention.

A.2 Trace accounting. Grouping $\Sigma_{\rho \in F_N} e^{2\pi i \rho}$ by denominator: the $q = 1$ vertices contribute $e^0 + e^{2\pi i} = 2 = c_1(1) + 1 = \mu(1) + 1$; each $q \geq 2$ contributes the Ramanujan sum $c_q(1) = \mu(q)$. Total: $M(N) + 1$ (Theorem 3.3). Under the alternative half-open convention $F_N \cap [0,1)$ (drop $1/1$), the trace is $M(N)$ and m decreases by 1; nothing downstream changes except the additive constants inside

Corollary 3.5's and Theorem 4.1's bookkeeping (replace `M(N)+1` by `M(N)` throughout - the `±1` absorptions are identical). The companion operator paper (Paper 1, DOI 10.5281/zenodo.21231873) states the same trace and uses the same closed convention.

A.3 Machine verification [measured]. The Version 2 release package contains the original bc009_commutator_identity_check.py as provenance, the bounded replay model/check_saturation_identity.py, the captured environment, and a SHA-256 inventory. The replay uses NumPy 2.4.6 and float64. Dense pseudoinverse checks are intentionally limited to `N=8,16`; the projection form of Theorem 3.1 and the Mertens trace are checked at `N=8,16,32,64,128,256,512`.

N	m(N)	M(N)+1	projection identity residual	D(N) from trace
8	23	-1	0.00e+00	1.8903591682e-03
16	81	0	0.00e+00	3.19e-33 (floating zero)
32	325	-3	0.00e+00	8.5207100592e-05
64	1261	0	0.00e+00	2.03e-33 (floating zero)
128	5023	-1	2.22e-16	3.9634523715e-08
256	19949	0	0.00e+00	0.00e+00
512	79853	-3	1.11e-16	1.4114322479e-09

For the dense checks, the identity residuals are `1.89e-15` at `N=8` and `8.66e-15` at `N=16`; the pseudoinverse and trace-form deficits agree to `1.91e-15` and `8.66e-15`, respectively. These are implementation checks, not evidence for an asymptotic rate or for RH. Exact commands, captured outputs, environment details, and hashes are included in reproducibility/final-preprint-replay/ inside the release bundle.

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